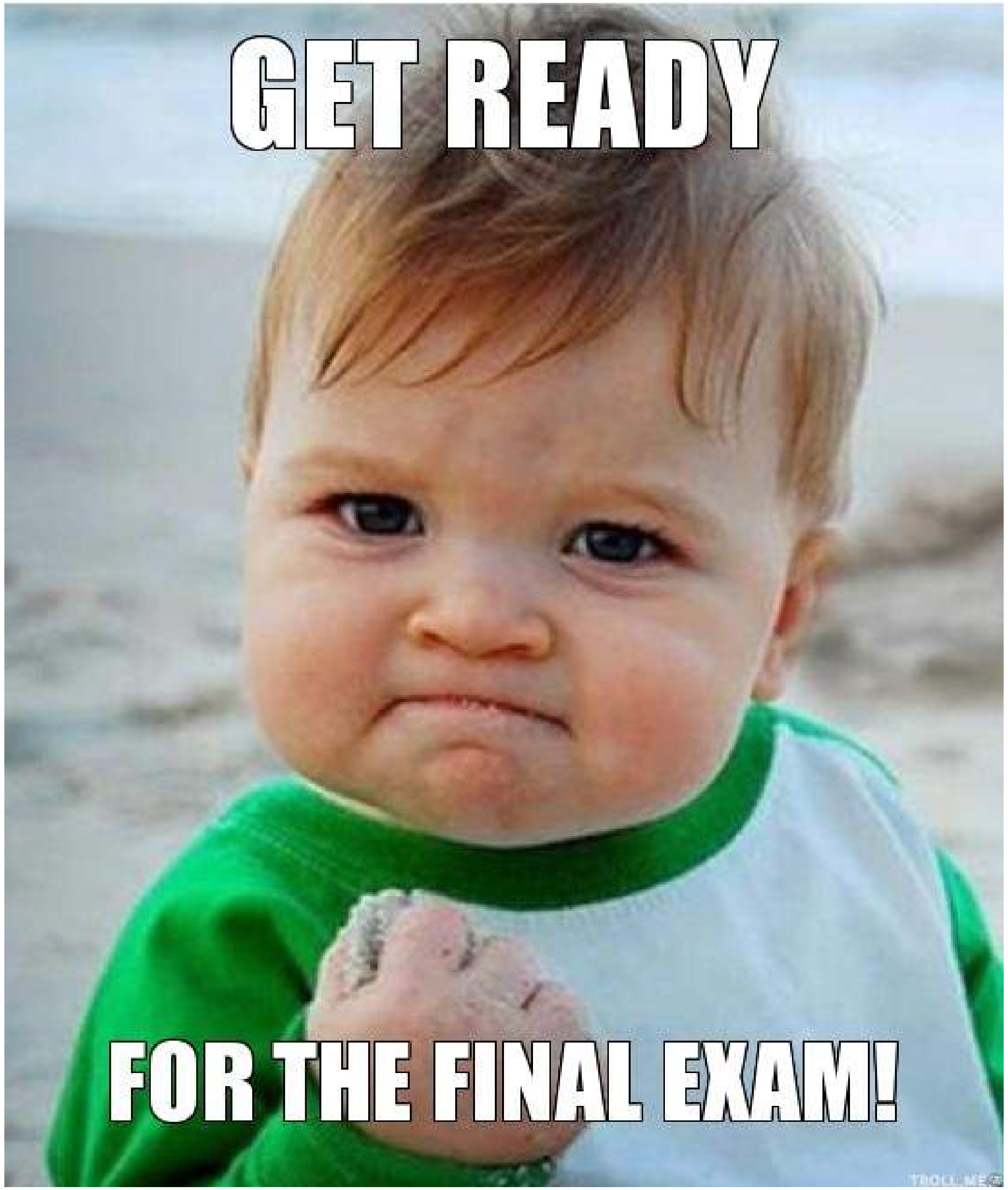
GET READY
FOR THE FINAL EXAM!

TP2 Exam 2020-21

- There are 4 questions:
  - 3 questions on Heat Transfer $\simeq 65\%$
  - 1 question on Fluid Mechanics
- You are required to answer **ALL** questions
- Formula sheet will be provided (a copy is available on Blackboard). Check Ronny's one too

Revision Materials

- **Lecture notes** (concepts and theory)
- **Lecture slides** (concepts and theory)
- Problem classes: slides for problem classes usually contain examples from lectures or from problems sheets;
- Solve the **Examples in lectures** first. Solutions are provided.
- Then solve the **Problem Sheets**. Proposed Solutions are provided.
- Finally solve questions from **past exam papers**.

Change in content

- Content was reduced in academic year 2018-19 and again in 2020-21
- When working on past exam papers, you may encounter questions on topics that are no longer part of the course:
  1. Correlations for flow boiling heat transfer
  2. Cooling towers
  3. Radiative heat transfer ~~from gases~~

Boiling

- Describe and apply conditions for bubble nucleation.
- Describe a typical pool boiling curve and the different regimes of pool boiling; understand and apply suitable heat transfer correlations for various boiling regimes.
- Understand the relationship between two-phase flow patterns and heat transfer coefficient in forced convection boiling in an upward tube heated uniformly along its length.
- Know the limiting factors in design of boiling equipment

Condensation (1)

- Nusselt theory for laminar film condensation: understand the assumptions and derive equations for condensate film thickness and local film heat transfer coefficient; apply Nusselt solutions to laminar film condensation on vertical plate, vertical and horizontal tubes. Know condensate film Reynolds number.
- Shear-controlled laminar film condensation: understand the assumptions and derive equations for condensate film thickness and local film heat transfer coefficient; apply the derived relations to laminar film condensation on vertical plate and vertical tubes.

Condensation (2)

- Equivalent laminar film (ELF) model: understand the concept of ELF model and apply it to the analysis of the effects of condensation rate on interfacial shear stress and heat transfer. Perform overall heat balance analysis for single component condensation.
- Condensation in the presence of a non-condensable component: apply the ELF model to the analysis of the effect of a non-condensable component on condensation; use the ELF model to calculate interfacial temperature and local heat flux for a given condensation rate.

Air-Water systems

- Basic definitions: humidity, relative humidity, Dew point, wet-bulb temperature.
- Understand the relationship between humidity, dry-bulb and wet-bulb temperatures.
- Use of psychrometric chart.
- Mass and energy balances for humidification, dehumidification, adiabatic mixing, and drying of wet solids. Use psychrometric chart to solve problems in these processes (single or combined).

May 2020 exam

Question 2 **[25 marks]**

cooling and dehumudifying

(a) Air at 1 atm, 34°C and 60% relative humidity needs to be brought to the "comfort" conditions of 24°C and 50% relative humidity using cooling and heating coils. The mass flow rate of the dry air is 100 kg h^{-1}.

(i) Draw a sketch of the required air-conditioning process in a simplified psychrometric chart.

[5 marks]

Using the provided psychrometric chart, estimate the following amounts:

(ii) The condensate mass flow rate collected.

[5 marks]

(iii) The heat rate in the cooling coil.

[5 marks]

(iv) The heat rate in the heating coil.

[5 marks]

Question 2 [25 marks]

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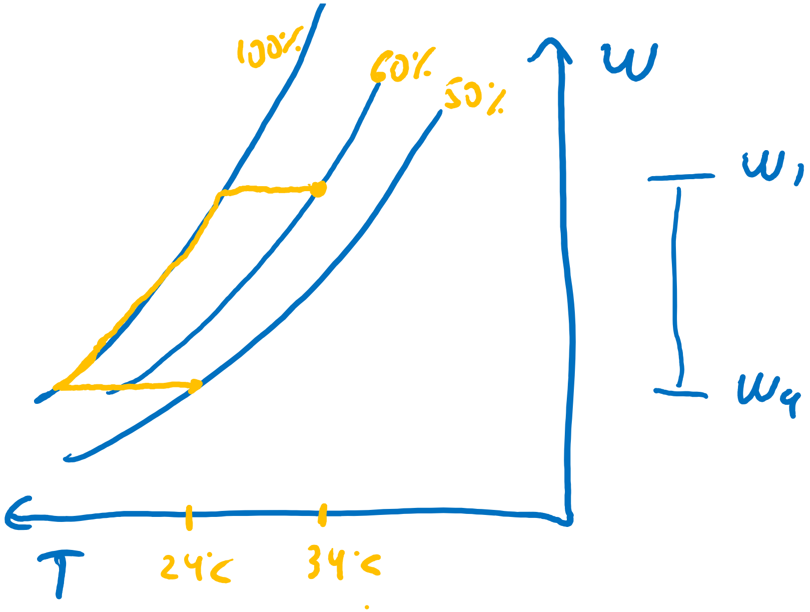
[5 marks]

i)



ii) total condensate $(w_1 - w_4)\,m_{a1} = 1.1$ Kg water h^{-1}

iii) Assuming $\dot{m}_w h_w$ small $(\dot{m}_w \ll \dot{m}_a)$ $\Rightarrow Q_c \Rightarrow$ graph $\Rightarrow 86.5 - 36.5 =$

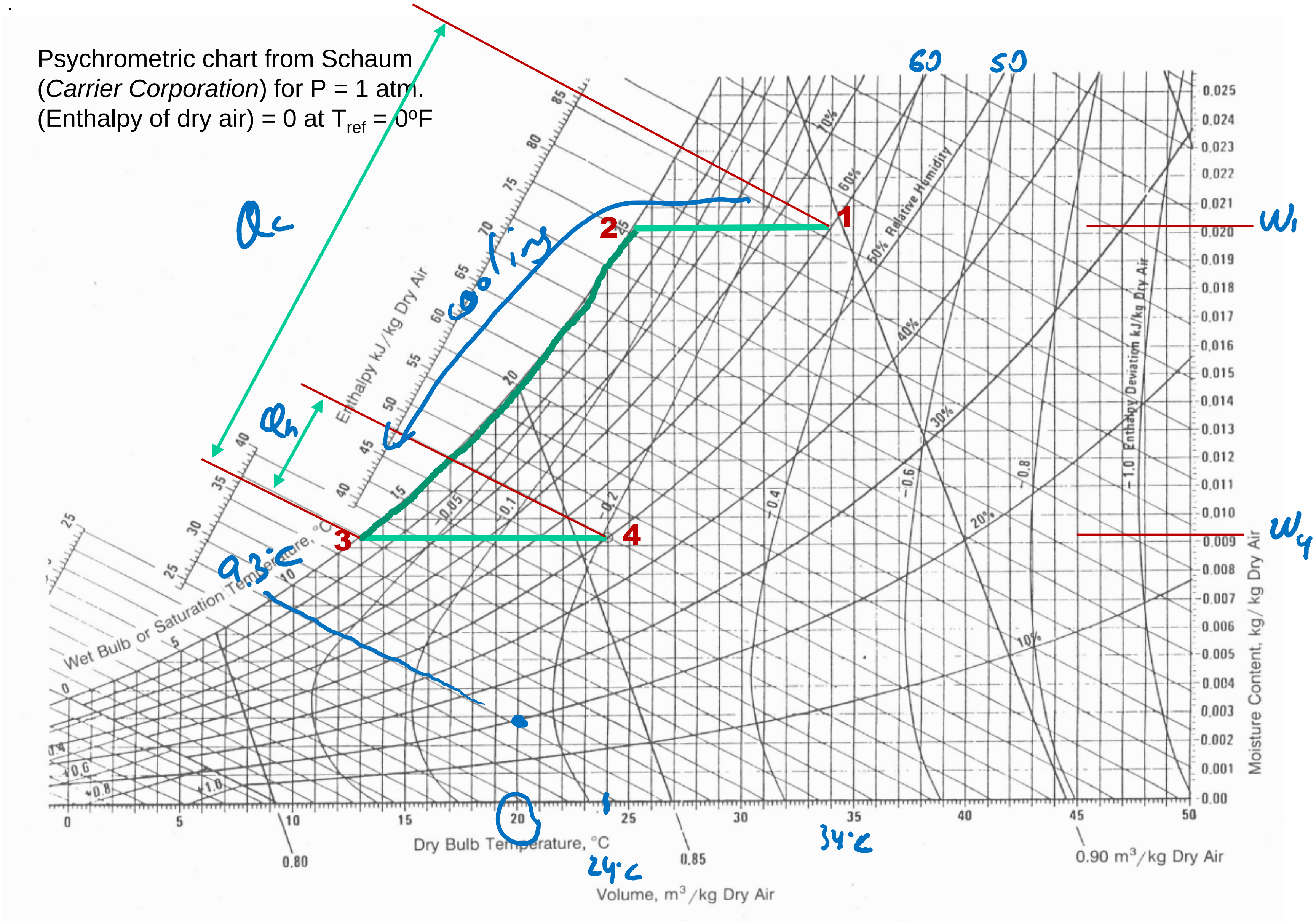
h_w small

$= 50 \dfrac{KJ}{Kg \text{ dry air}}$

In 100 Kg dry air $h^{-1} \rightarrow 5000$ KJ h^{-1}

iv) $Q_h \Rightarrow$ graph $\rightarrow 11.5$ KJ/Kg dry air

$\times 100 \rightarrow 1150$ KJ h^{-1}



Psychrometric chart from Schaum
(Carrier Corporation) for P = 1 atm.
(Enthalpy of dry air) = 0 at T_ref = 0°F
Enthalpy kJ/kg Dry Air
Wet Bulb or Saturation Temperature, °C
Dry Bulb Temperature, °C
Volume, m³/kg Dry Air
Moisture Content, kg/kg Dry Air
Enthalpy Deviation kJ/kg Dry Air
50% Relative Humidity
0.80
0.85
0.90 m³/kg Dry Air
1
2
3
4
Qc
Qh
cooling
w1
w4
9.3°C
24°C
34°C
60
50

(b) Inside a duct, warm water at 40°C and cold atmospheric air at 20°C and 20% relative humidity are encountered in counterflow mode, as shown in Figure 2.

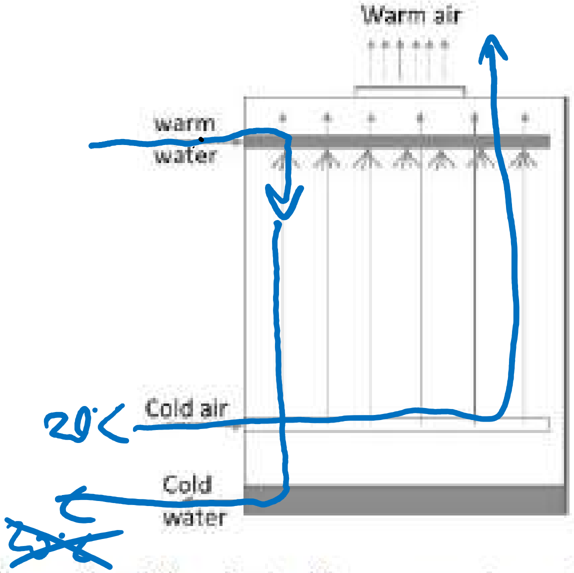


Figure 2 – Schematic of the duct with warm water and cold air in counterflow.

Using the provided psychrometric chart, estimate the minimum temperature that the cold water could achieve at the exit.

[5 marks]

END OF QUESTION 2

(b) Inside a duct, warm water at 40°C and cold atmospheric air at 20°C and 20% relative humidity are encountered in counterflow mode, as shown in Figure 2.

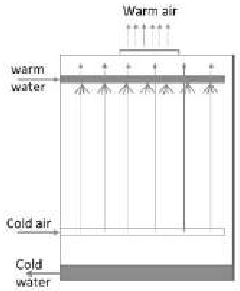


Figure 2 – Schematic of the duct with warm water and cold air in counterflow.

Using the provided psychrometric chart, estimate the minimum temperature that the cold water could achieve at the exit.

[5 marks]

END OF QUESTION 2

$T_{WB} \Rightarrow 9.3\ ^\circ C$